

CBSE Class 10 Mathematics — Constructions — Solutions

1. 1. To draw a pair of tangents to a circle which are inclined to each other at an angle of 60° , it is required to draw tangents at the endpoints of those two radii of the circle, the angle between which is:

Answer: (C)

Solution:

1. The angle between the tangents and the angle between the radii at the points of contact are supplementary.
2. Let the angle between the radii be θ .
3. $\theta + 60^\circ = 180^\circ$
4. $\theta = 120^\circ$

Derivation:

$$\theta = 180^\circ - 60^\circ = 120^\circ$$

Constructions — Construction of tangents

2. 2. If a point P is 10 cm away from the centre of a circle of radius 6 cm, the length of the tangent drawn from P to the circle is:

Answer: (B)

Solution:

1. The radius is perpendicular to the tangent at the point of contact.
2. This forms a right-angled triangle with the hypotenuse as the distance from the centre.
3. Use Pythagoras theorem: $d^2 = r^2 + t^2$
4. $10^2 = 6^2 + t^2$

Derivation:

$$t = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8 \text{ cm}$$

Constructions — Construction of tangents

3. 3. How many tangents can be drawn from an external point to a circle?

Answer: (B)

Solution:

1. A tangent touches the circle at exactly one point.
2. From a point outside a circle, exactly two distinct tangents can be drawn to the circle.

Derivation:

$$\text{Number of tangents} = 2$$

Constructions — Construction of tangents

4. 4. To divide a line segment AB in the ratio $3 : 4$, we draw a ray AX such that $\angle BAX$ is acute. Then we mark points on AX at equal distances. The minimum number of points to be marked is:

Answer: (C)

Solution:

1. To divide a line segment in ratio $m : n$, we need to mark $m + n$ points on the ray.
2. Here $m = 3$ and $n = 4$.
3. Total points $= 3 + 4 = 7$

Derivation:

$$3 + 4 = 7$$

Constructions — Construction of tangents

5. 5. The angle between a tangent to a circle and the radius passing through the point of contact is:

Answer: (C)

Solution:

1. By the theorem of tangents, the tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. Perpendicular means 90° .

Derivation:

$$\text{Angle} = 90^\circ$$

Constructions — Construction of tangents

6. 6. Assertion (A): The length of the tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.

Answer: (D)

Solution:

1. The length of a tangent is independent of the radius length.
2. The theorem states that a tangent is perpendicular to the radius at the point of contact.

Derivation:

$$A : \text{False}, R : \text{True}$$

Constructions — construction of tangents

7. **7.** Assertion (A): To construct a pair of tangents to a circle from an external point, the angle between the tangents is supplementary to the angle subtended by the line segment joining the points of contact at the center. Reason (R): The sum of opposite angles in a cyclic quadrilateral is 180° .

Answer: (A)

Solution:

1. The quadrilateral formed by the center, two contact points, and the external point has two right angles.
2. The sum of angles is 360° , so the remaining two must sum to 180° .

Derivation:

$A : \text{True}, R : \text{True}, R \text{ explains } A$

Constructions — construction of tangents

8. **8.** Assertion (A): It is impossible to construct a tangent to a circle from a point lying inside the circle. Reason (R): A tangent touches the circle at only one point.

Answer: (A)

Solution:

1. A line passing through an interior point will always be a secant, not a tangent.
2. A tangent by definition meets the circle at exactly one point on the circumference.

Derivation:

$A : \text{True}, R : \text{True}, R \text{ explains } A$

Constructions — construction of tangents

9. **9.** Assertion (A): The number of tangents that can be drawn from a point on the circle to the circle is exactly two. Reason (R): A point on the circle lies on the circumference.

Answer: (D)

Solution:

1. Only one tangent can be drawn from a point on the circle.
2. The reason is a true statement about the location of the point.

Derivation:

$A : \text{False}, R : \text{True}$

Constructions — construction of tangents

10. **10.** Assertion (A): When constructing tangents to a circle from an external point P , we bisect the line segment OP where O is the center. Reason (R): The perpendicular bisector of OP helps in finding the midpoint which acts as the center of a circle passing through O and P .

Answer: (A)

Solution:

1. To find points of contact, we draw a circle with diameter OP .
2. The midpoint is found by constructing the perpendicular bisector of OP .

Derivation:

$$A : \text{True}, R : \text{True}, R \text{ explains } A$$

Constructions — construction of tangents

- 11. 11.** Draw a circle of radius 3 cm. Take two points P and Q on one of its extended diameter each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points P and Q .

Answer:

Two pairsoftangentsareconstructedfromPandQ.

Solution:

1. Draw a circle of radius 3 cm with centre O . Extend the diameter on both sides to points P and Q such that $OP = OQ = 7$ cm.
2. Bisect OP at M and OQ at N . Draw circles with centers M and N and radius MP and NQ respectively, intersecting the original circle at the required points. Join the points of intersection to P and Q respectively.

Derivation:

Constructioninvolvesperpendicularbisectormethodtofindthemidpointsofthelinesconnectingtheea

Constructions — construction of tangents

- 12. 12.** Construct a tangent to a circle of radius 4 cm from a point on the concentric circle of radius 6 cm and measure its length.

Answer:

Thelengthofthetangentisapproximately4.47 cm.

Solution:

1. Draw two concentric circles with centre O and radii 4 cm and 6 cm. Let P be a point on the outer circle.
2. Join OP . Bisect OP at M . Draw a circle with centre M and radius MO to intersect the inner circle at A and B . Join PA and PB , which are the required tangents.

Derivation:

$$L = \sqrt{R^2 - r^2} = \sqrt{6^2 - 4^2} = \sqrt{36 - 16} = \sqrt{20} \approx 4.47 \text{ cm.}$$

Constructions — construction of tangents

- 13. 13.** Draw a line segment AB of length 8 cm. Taking A as centre, draw a circle of radius 4 cm and taking B as centre, draw another circle of radius 3 cm. Construct tangents to each circle from the centre of the other circle.

Answer:

TangentsconstructedfromBtocircleAandfromAtocircleB.

Solution:

1. Draw segment $AB = 8$ cm. Construct circles of radii 4 cm and 3 cm at A and B .
2. Bisect AB at M . Draw a circle with centre M and radius MA (or MB). This circle intersects the circle at A at two points and the circle at B at two points. Join these points to B and A respectively.

Derivation:

The construction uses the property that the angle in a semicircle is 90° , thus defining the points of tangency.

Constructions — construction of tangents

14. 14. Draw a circle of radius 3.5 cm. Construct a pair of tangents to this circle which are inclined to each other at an angle of 60° .

Answer:

The tangents are constructed such that the angle between radii at the point of contact is 120° .

Solution:

1. Draw two radii OA and OB such that $\angle AOB = 120^\circ$ (since $180^\circ - 60^\circ = 120^\circ$).
2. Construct perpendiculars at A and B to radii OA and OB . These perpendiculars meet at point P , where $\angle APB = 60^\circ$.

Derivation:

The quadrilateral formed by the center, the two points of contact, and the external point has an angle sum of 360° .

Constructions — construction of tangents

15. 15. Draw a circle of radius 5 cm. Mark a point P at a distance of 8 cm from the centre. Construct the two tangents from P to the circle.

Answer:

Two tangents from P are constructed.

Solution:

1. Draw a circle with centre O and radius 5 cm. Mark point P such that $OP = 8$ cm.
2. Bisect OP at M . Draw a circle with centre M and radius MO . Mark the intersection points A and B on the original circle. Join PA and PB .

Derivation:

The construction relies on the property that the tangent is perpendicular to the radius at the point of contact.

Constructions — construction of tangents

16. **16.** Draw a circle of radius 3 cm. Take two points P and Q on one of its extended diameters each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points P and Q .

Answer:

Two pair of tangents constructed from points at distance 7 cm

Solution:

1. Draw a circle with center O and radius 3 cm.
2. Draw a diameter and extend it on both sides to locate points P and Q such that $OP = OQ = 7$ cm.
3. Bisect OP and OQ to find midpoints M_1 and M_2 .
4. With M_1 and M_2 as centers, draw circles of radius M_1O to intersect the original circle at two points each. Join P and Q to these intersection points.

Derivation:

Construction follows the geometric property : angle in a semicircle is 90° .

Constructions — Construction of tangents

17. **17.** Construct a tangent to a circle of radius 4 cm from a point on the concentric circle of radius 6 cm and measure its length.

Answer:

$$\sqrt{20} \approx 4.47 \text{ cm}$$

Solution:

1. Draw two concentric circles with radii 4 cm and 6 cm with center O .
2. Mark a point P on the circumference of the larger circle.
3. Join OP and bisect it to find midpoint M .
4. Draw a circle with center M and radius MO to cut the inner circle at A and B . PA and PB are the required tangents.

Derivation:

$$\text{Length } L = \sqrt{6^2 - 4^2} = \sqrt{36 - 16} = \sqrt{20} = 2\sqrt{5}$$

Constructions — Construction of tangents

18. **18.** Draw a line segment AB of length 8 cm. Taking A as center, draw a circle of radius 4 cm and taking B as center, draw another circle of radius 3 cm. Construct tangents to each circle from the center of the other circle.

Answer:

Two pair of tangents constructed

Solution:

1. Draw segment $AB = 8$ cm. Draw circle C_1 (center A , $r = 4$) and C_2 (center B , $r = 3$).

2. Bisect AB to get midpoint M .
3. Draw a circle with center M and radius MA (or MB).
4. The points where this circle intersects C_1 and C_2 are the points of tangency. Join B to intersection points on C_1 and A to intersection points on C_2 .

Derivation:

Tangents are constructed by finding the intersection of circles centered at the midpoint of the line segment

Constructions — Construction of tangents

- 19. 19.** Draw a circle of radius 3.5 cm. Construct a pair of tangents to this circle which are inclined to each other at an angle of 60° .

Answer:

Construction of tangents at 120° central angle

Solution:

1. Draw a circle of radius 3.5 cm with center O .
2. Draw a radius OA . Draw another radius OB such that $\angle AOB = 120^\circ$ (since $180^\circ - 60^\circ = 120^\circ$).
3. At point A , draw a perpendicular to OA .
4. At point B , draw a perpendicular to OB . The intersection point P gives the required tangents PA and PB .

Derivation:

Sum of opposite angles in the quadrilateral $OAPB$ is 180° . Thus, if $\angle APB = 60^\circ$, $\angle AOB = 120^\circ$.

Constructions — Construction of tangents

- 20. 20.** Construct a triangle with sides 5 cm, 6 cm and 7 cm. Then construct a circle touching the side of 6 cm at its midpoint and passing through the other vertex.

Answer:

Construction of a circle through specific vertices tangent to a side

Solution:

1. Construct $\triangle ABC$ with $AB = 5$, $BC = 6$, $AC = 7$.
2. Locate midpoint M of BC .
3. Draw a perpendicular to BC at M .
4. Draw the perpendicular bisector of AM . The intersection of these two perpendiculars is the center O . Draw the circle with radius OM .

Derivation:

The center must be equidistant from the point of tangency M and vertices A and C .

Constructions — Construction of tangents

- 21. 21.** Draw a circle of radius 3 cm. Take two points P and Q on one of its extended diameters each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points P and Q .

Answer:

Two pairs of tangents constructed from points P and Q , each pair having equal length of $\sqrt{7^2 - 3^2} = \sqrt{40}$ cm.

Solution:

1. Draw a circle with center O and radius 3 cm.
2. Draw a diameter and extend it on both sides to locate P and Q such that $OP = OQ = 7$ cm.
3. Bisect OP and OQ to find midpoints M and N .
4. Using M as center and MO as radius, draw a circle to intersect the original circle at points A and B .
5. Using N as center and NO as radius, draw a circle to intersect the original circle at points C and D .
6. Join PA , PB , QC , and QD to get the required tangents.

Derivation:

$$\text{Length of tangent} = \sqrt{d^2 - r^2} = \sqrt{7^2 - 3^2} = \sqrt{49 - 9} = \sqrt{40} \text{ cm.}$$

Constructions — Construction of tangents to a circle

- 22. 22.** Construct a triangle ABC with side $BC = 6$ cm, $AB = 5$ cm and $\angle ABC = 60^\circ$. Then construct a triangle whose sides are $\frac{3}{4}$ of the corresponding sides of the triangle ABC . Finally, construct a tangent to the circumcircle of triangle ABC at vertex A .

Answer:

A scaled triangle $A'BC'$ and a tangent line touching the circumcircle at A .

Solution:

1. Construct triangle ABC with given dimensions.
2. Draw an acute angle at B and divide the line into 4 segments to construct the similar triangle $A'BC'$.
3. Find the circumcenter O by drawing perpendicular bisectors of AB and BC .
4. Draw the circumcircle with center O passing through A , B , and C .
5. Construct a line perpendicular to radius OA at point A to get the tangent.

Derivation:

The tangent at A is perpendicular to the radius OA , hence $\angle OAT = 90^\circ$.

Constructions — Construction of tangents and similar triangles

- 23. 23.** Draw a pair of tangents to a circle of radius 4 cm, which are inclined to each other at an angle of 60° . Justify the construction.

Answer:

Tangents constructed such that the angle between them is 60° .

Solution:

1. Draw a circle of radius 4 cm with center O.
2. Draw a radius OA.
3. Draw a radius OB such that $\angle AOB = 180^\circ - 60^\circ = 120^\circ$.
4. At point A, construct a line perpendicular to OA.
5. At point B, construct a line perpendicular to OB.
6. The point where these two lines intersect is P. $\angle APB$ will be 60° .

Derivation:

In quadrilateral OAPB, $\angle OAP = 90^\circ$, $\angle OBP = 90^\circ$, and $\angle AOB = 120^\circ$. Thus $\angle APB = 360^\circ - (90^\circ + 90^\circ + 120^\circ) = 60^\circ$.

Constructions — Construction of tangents

- 24. 24.** Draw two concentric circles of radii 3 cm and 5 cm. Construct a tangent to the smaller circle from a point on the larger circle and measure its length. Verify the measurement by calculation.

Answer:

Length of tangent is 4 cm.

Solution:

1. Draw two concentric circles with center O and radii 3 cm and 5 cm.
2. Mark a point P on the circumference of the larger circle.
3. Join OP (length 5 cm).
4. Bisect OP at M.
5. Draw a circle with center M and radius MO, cutting the smaller circle at A and B.
6. Join PA and PB, which are the required tangents.

Derivation:

In right $\triangle OAP$, $OA = 3$ cm, $OP = 5$ cm. By Pythagoras theorem, $PA = \sqrt{OP^2 - OA^2} = \sqrt{5^2 - 3^2} = 4$ cm.

Constructions — Construction of tangents